

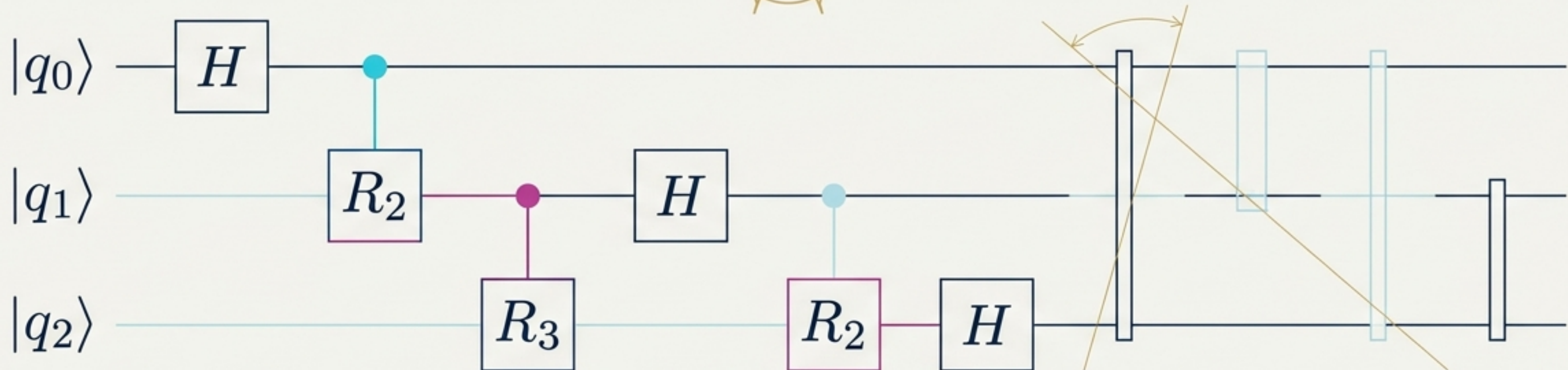
ENGINEERING THE QUANTUM STATE

LECTURE 6: QUANTUM FOURIER TRANSFORM (QFT)

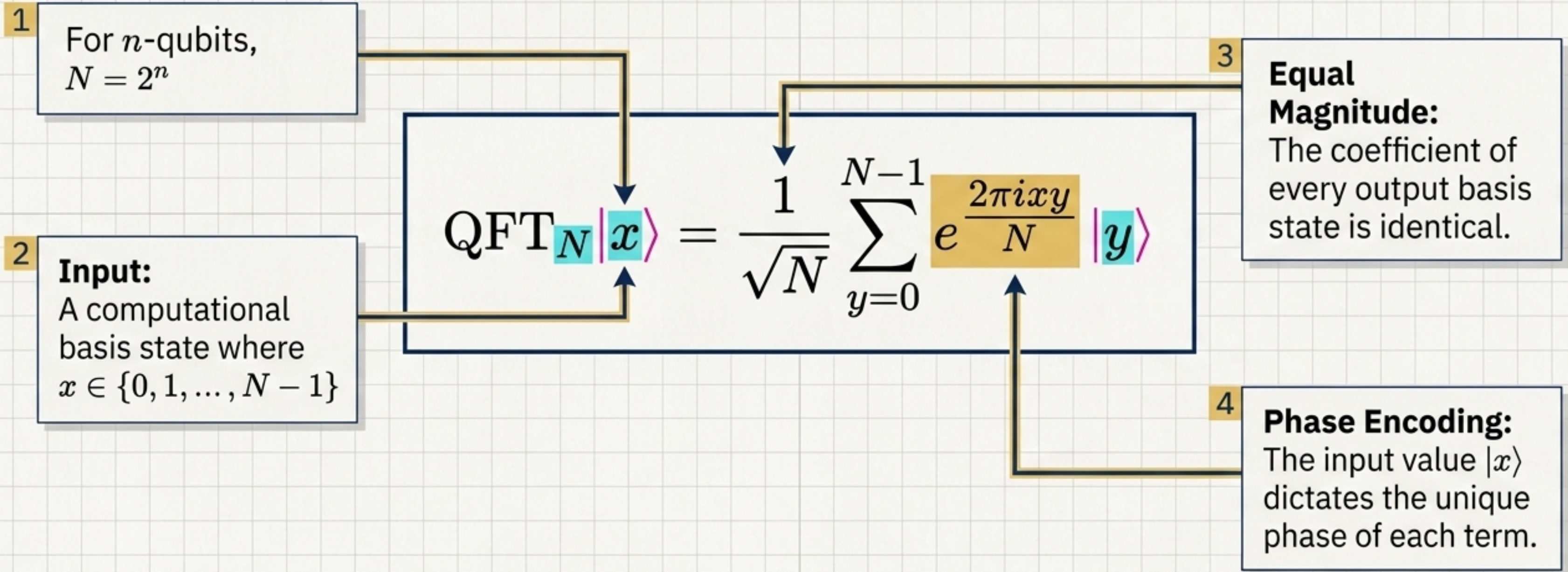
LECTURE 6: QUANTUM FOURIER TRANSFORM (QFT)

AND QUANTUM PHASE ESTIMATION (QPE)

SURVEY OF QUANTUM TECHNOLOGIES & APPLICATIONS (PH 40001) | ARIGHNA DEB, KIIT UNIVERSITY



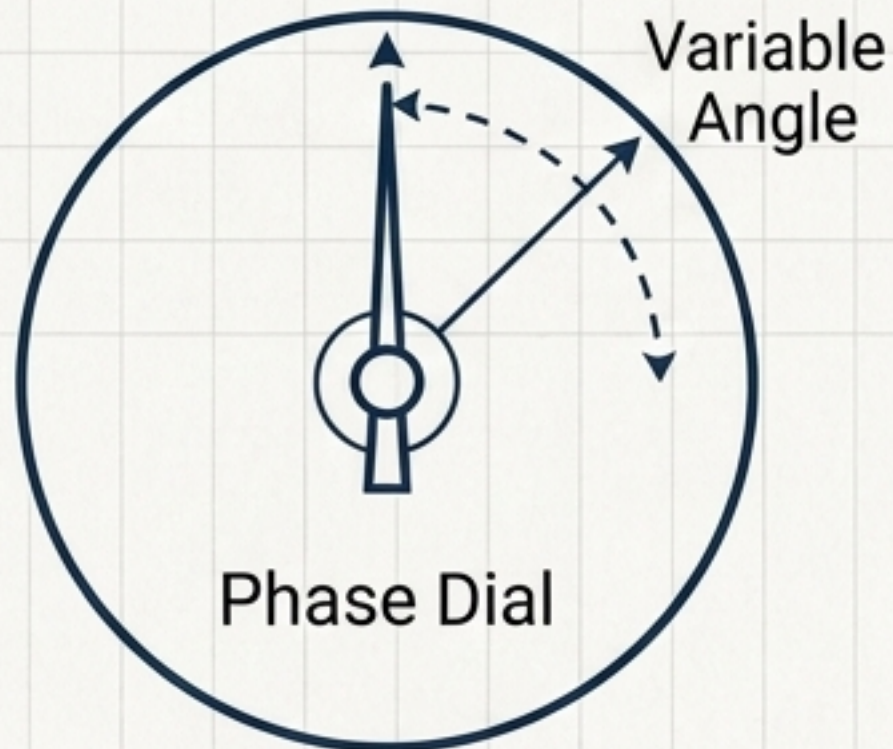
The QFT translates discrete computational states into phase-encoded amplitudes



Phase rotation gates (R_k) precisely tune the phase of the $|1\rangle$ state

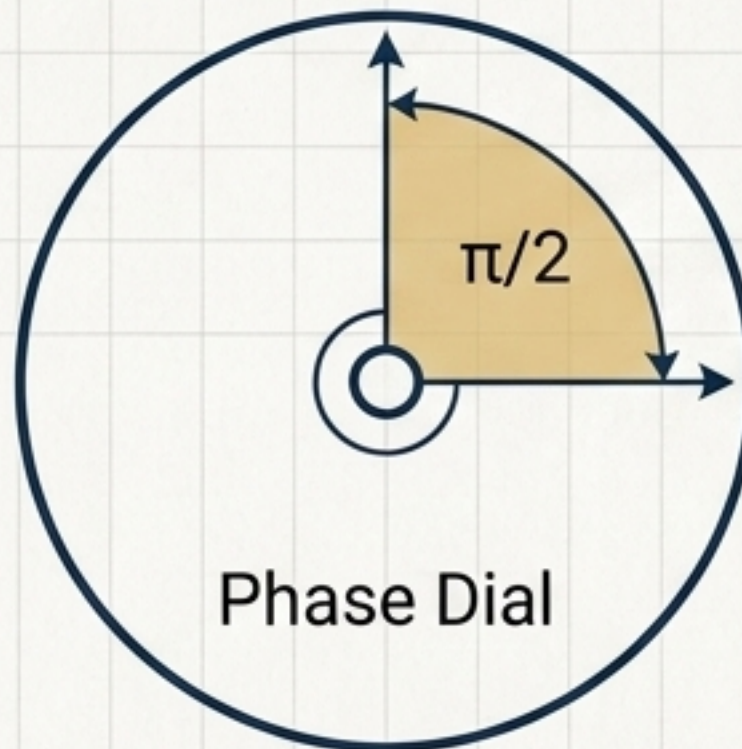
General Operator (R_k)

$$\begin{bmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^k} \end{bmatrix}$$



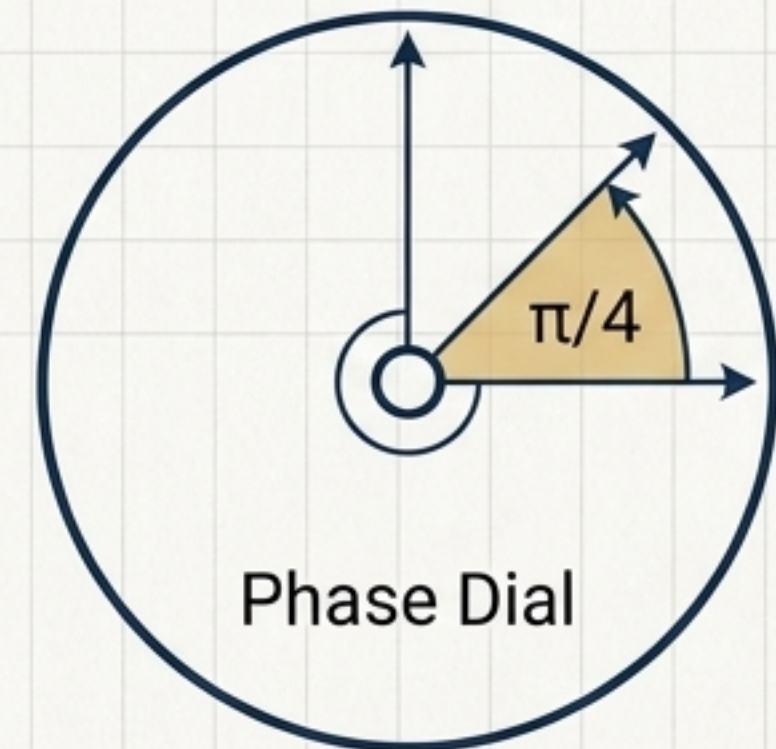
The S Gate ($k=2$)

$$R_2 = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$



The T Gate ($k=3$)

$$R_3 = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$$



At the 1-qubit scale, the QFT collapses into the familiar Hadamard gate

The Mathematical Proof

For $n=1$, $N=2$.

$$\text{QFT}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = H$$

The Circuit Equivalent

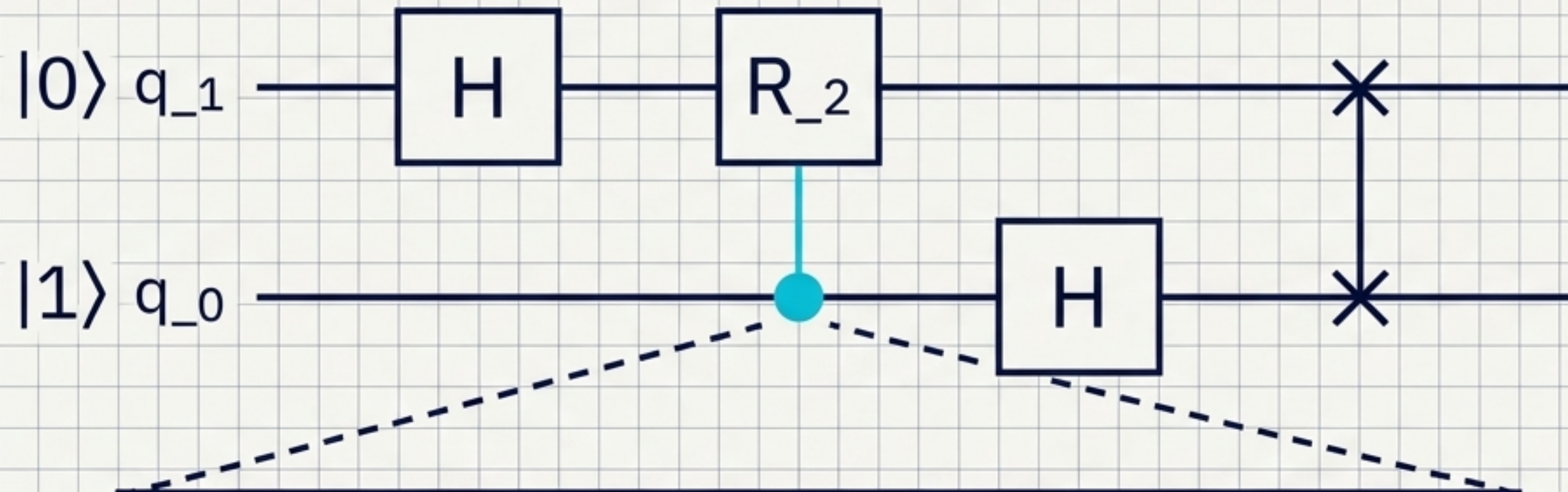


$$H|0\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

$$H|1\rangle = (|0\rangle - |1\rangle) / \sqrt{2}$$

The QFT is simply a generalization of the Hadamard transform across multiple dimensions.

Scaling to two qubits introduces controlled rotations and state swapping



CR_2 (Controlled- R_2) Mechanism

If control qubit is $|1\rangle$, target qubit receives an R_2 rotation.

$$\text{CR}_2 \left[\frac{(|01\rangle + |11\rangle)}{\sqrt{2}} \right] = \frac{(|01\rangle + i|11\rangle)}{\sqrt{2}}$$

The 2-qubit QFT yields a unique phase signature for every basis state

$$\text{QFT } |00\rangle = \frac{1}{2} [|00\rangle + |01\rangle + |10\rangle + |11\rangle]$$

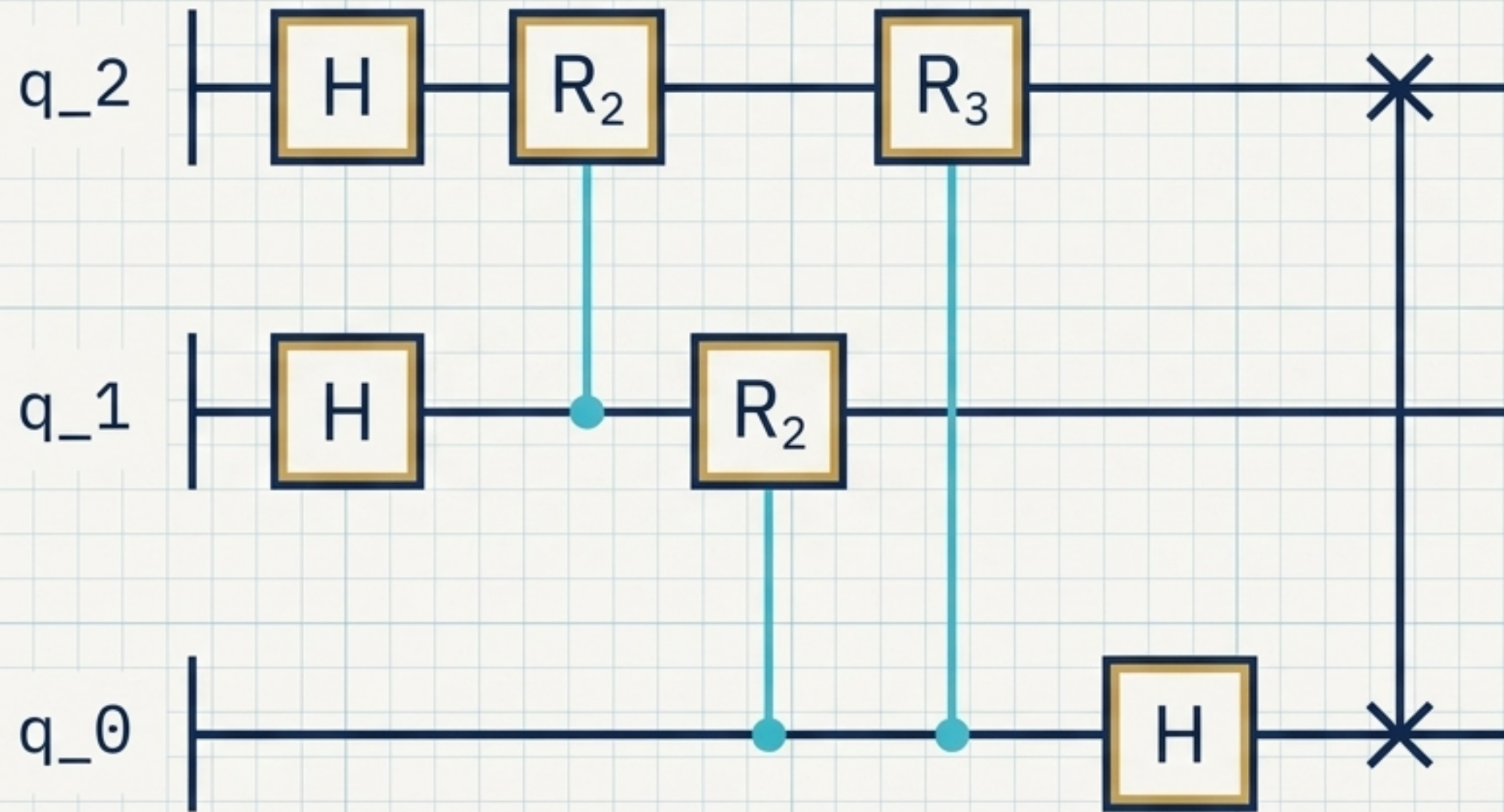
$$\text{QFT } |01\rangle = \frac{1}{2} [|00\rangle + i|01\rangle - |10\rangle - i|11\rangle]$$

$$\text{QFT } |10\rangle = \frac{1}{2} [|00\rangle - |01\rangle + |10\rangle - |11\rangle]$$

$$\text{QFT } |11\rangle = \frac{1}{2} [|00\rangle - i|01\rangle - |10\rangle + i|11\rangle]$$

Notice how the interference pattern (the unique combination of signs and imaginary components) distinctly tags the initial input state.

Generalizing to n-qubits reveals an $O(n^2)$ gate complexity

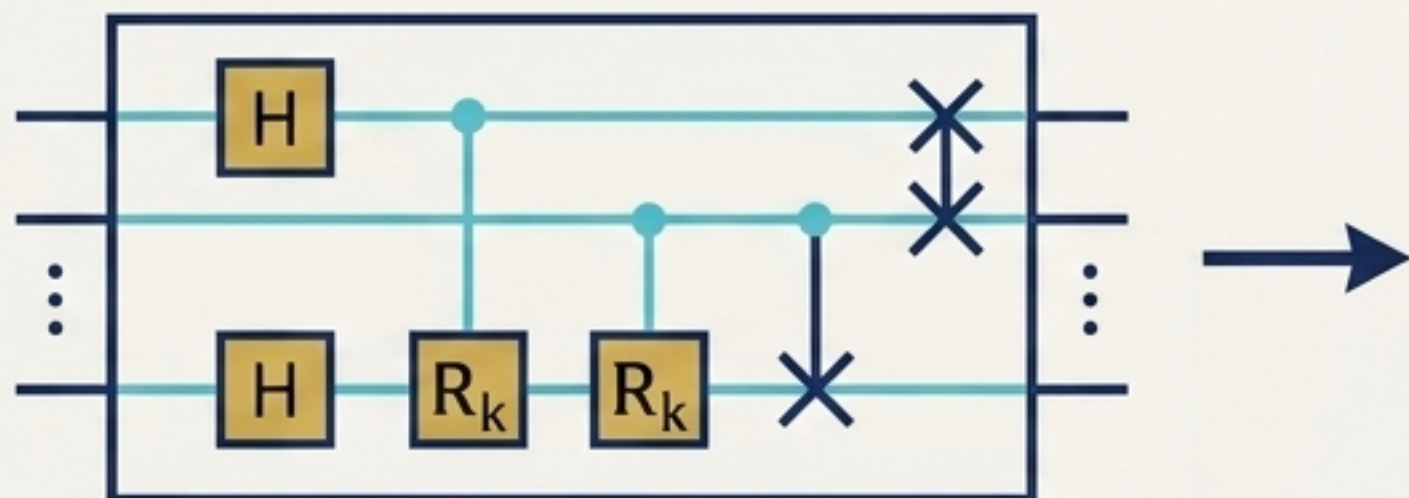


Scaling to n-qubits

- Exact QFT requires $n(n-1)/2$ controlled phase rotations.
- Total gate count scales quadratically to $O(n^2)$.

Inverse QFT (IQFT) unwinds the phase encoding to allow measurement

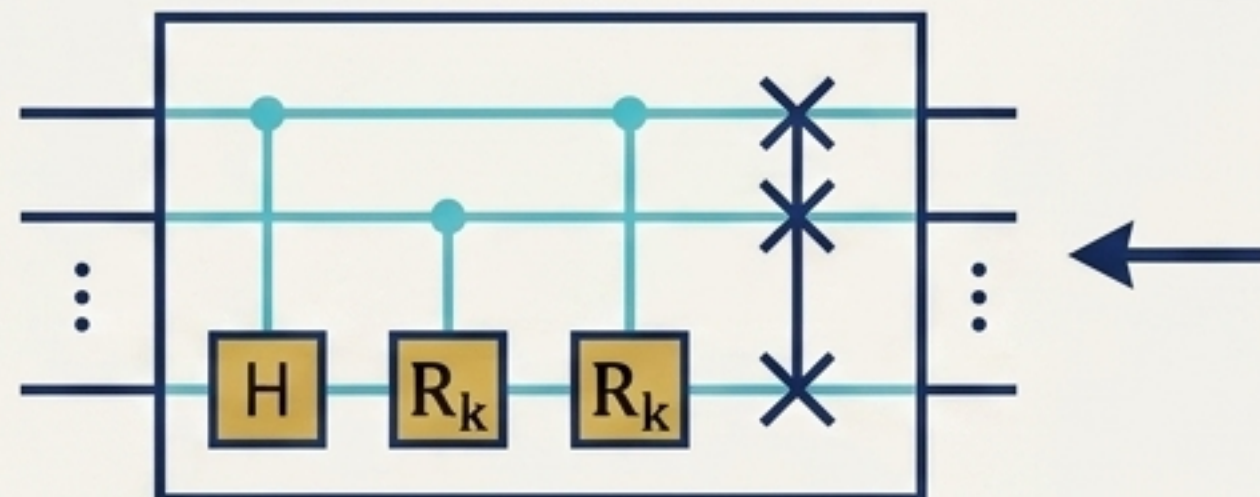
Standard QFT



mirror

Inverse QFT

$$\text{QFT}^\dagger = \text{QFT}^{-1}$$
$$\text{QFT}^\dagger \text{QFT} = \text{I}$$



3 Rules for Inversion Circuit

1. Reverse the exact gate order of QFT.
2. Replace each R_k phase rotation with its Hermitian conjugate $R_k^\dagger$.
3. Leave all **H** and SWAP gates exactly as they are.

Quantum Phase Estimation (QPE) is the engine behind quantum advantage

Let $\mathbf{U}$ be a unitary operator and $|\psi\rangle$ be one of its eigenstates.

$$\mathbf{U} |\psi\rangle = e^{2\pi i \phi} |\psi\rangle$$

The objective of QPE is to accurately estimate the unknown decimal phase ϕ .



Eigenvalue &
Energy Estimation



Hamiltonian
Simulation



Quantum Chemistry
& Drug Discovery

Phase Kickback forces the target eigenstate to reveal its phase to the control qubit

Transformation Timeline

Before CU gate

Start with a control qubit in superposition and an eigenstate $|\psi\rangle$.

$$|\psi_0\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + |1\rangle \right) |\psi\rangle$$

Apply Controlled-U (CU) Gate

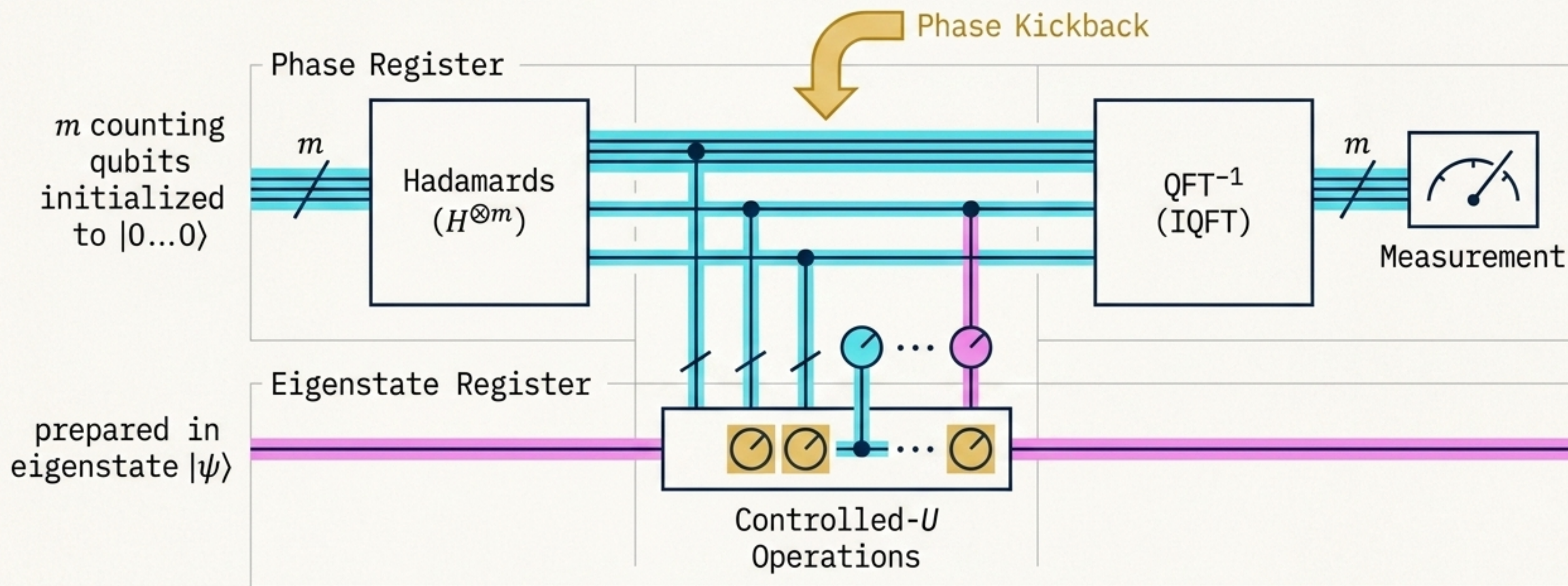
After CU gate

$$|\psi_1\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle |\psi\rangle + |1\rangle U|\psi\rangle \right)$$

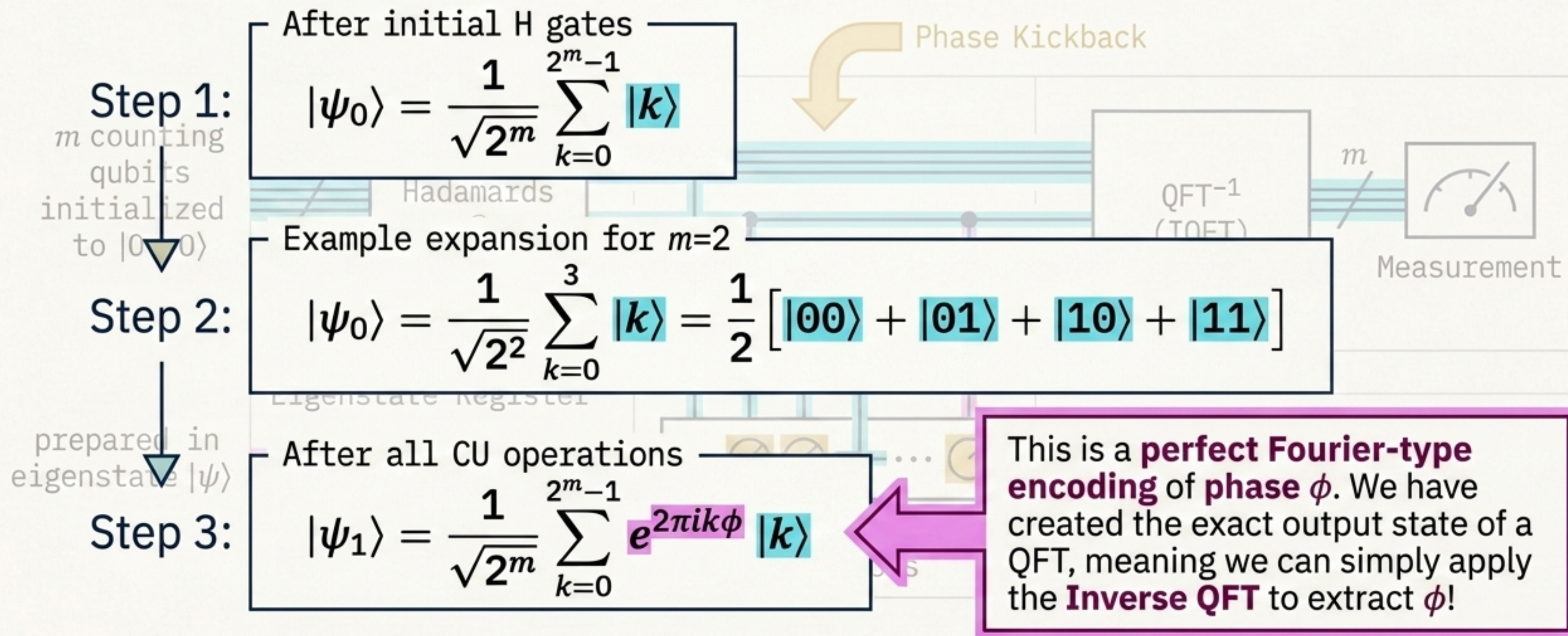
$$|\psi_1\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i \phi} |1\rangle \right) |\psi\rangle$$

Phase Kickback

The universal QPE architecture consists of two interconnected registers

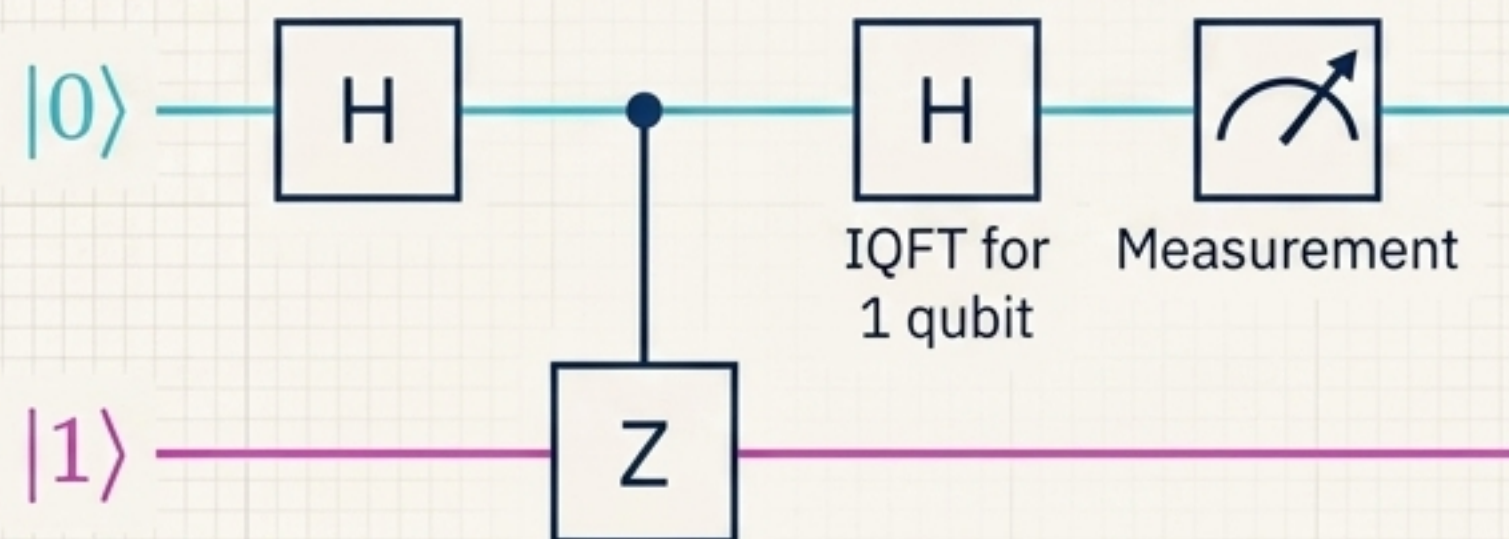


Tracking the systemic state evolution reveals a hidden Fourier sequence



A concrete example: 1-qubit phase estimation of the Pauli Z gate

The Circuit



1-qubit QPE circuit for $U=Z$ and eigenstate $|1\rangle$

The Mathematical Proof

Parameters: $U = Z$, Eigenstate = $|1\rangle$

$$Z|1\rangle = -|1\rangle = e^{i\pi}|1\rangle = e^{2\pi i(1/2)}|1\rangle$$

Therefore, expected $\phi = \frac{1}{2}$

Circuit Trace:

$$|\psi_0\rangle = |01\rangle$$

$$|\psi_1\rangle \text{ (After H)} = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) |1\rangle$$

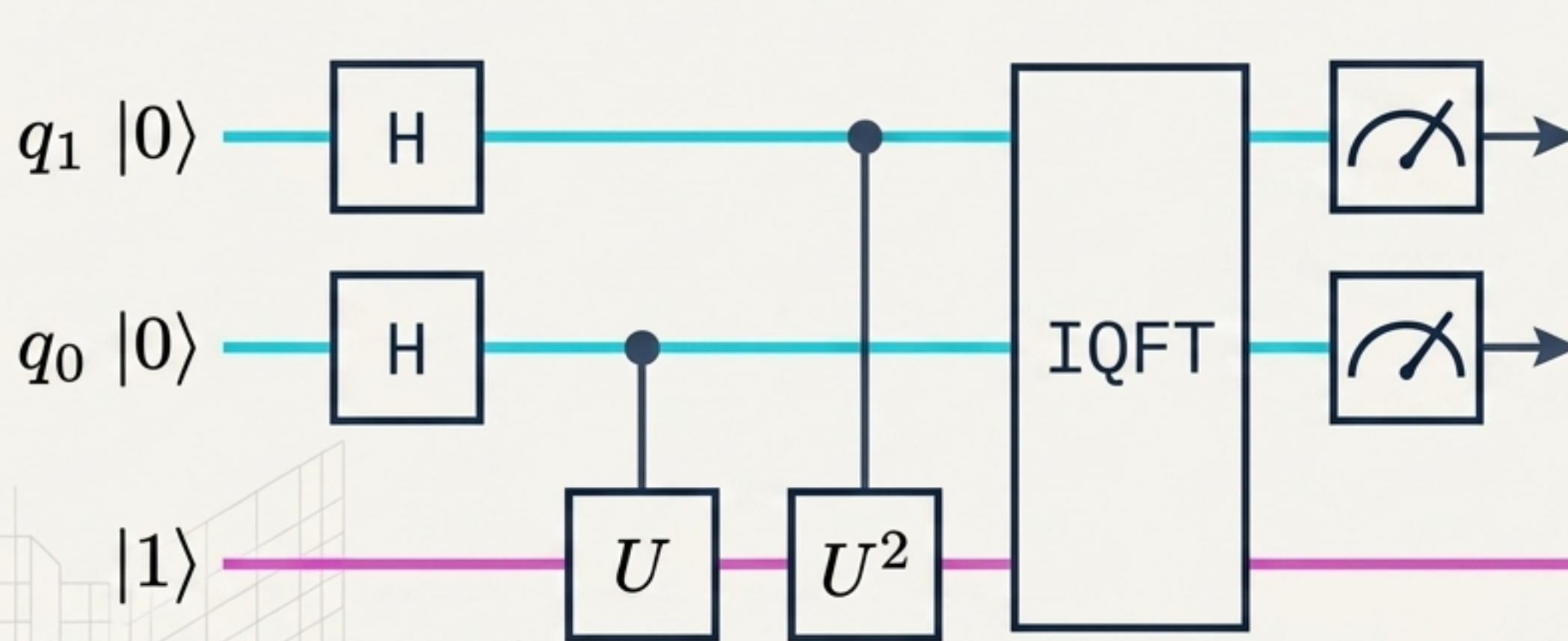
$$|\psi_2\rangle \text{ (After CZ)} = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) |1\rangle = |\rightarrow\rangle |1\rangle$$

$$|\psi_3\rangle \text{ (After IQFT/H)} = H|\rightarrow\rangle = |1\rangle$$

Conclusion:

Measuring the top qubit yields a binary 1, exactly representing $\phi = \frac{1}{2}$.

Higher precision requires more counting qubits and binary fraction readout



Parameters: $U = S$ gate, Eigenstate = $|1\rangle$.
Note that $U^2 = Z$.

The Mathematical Proof

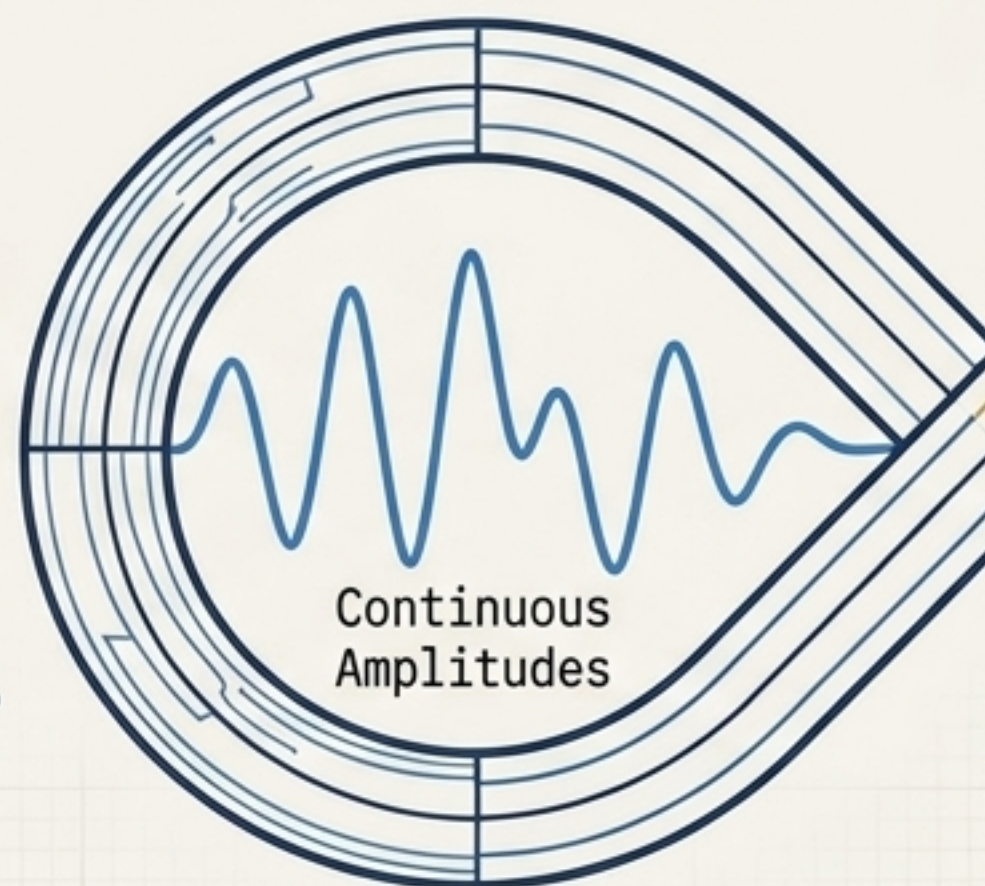
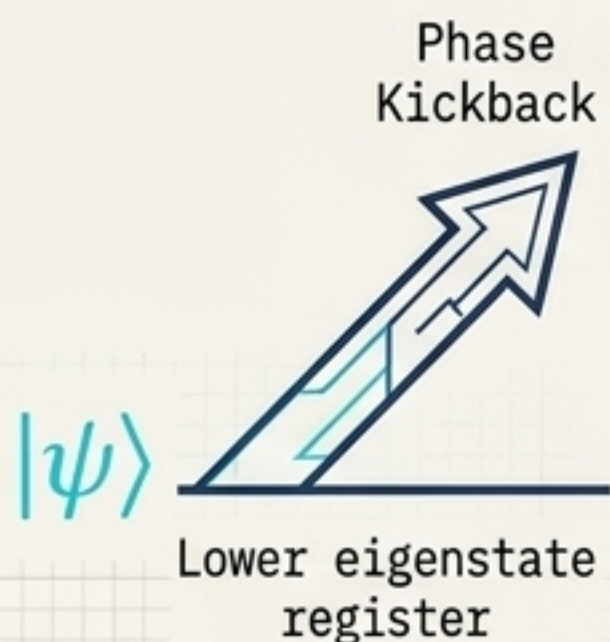
$$\begin{aligned} S|1\rangle &= i|1\rangle \\ &= e^{i\pi/2} |1\rangle \\ &= e^{2\pi i(1/4)} |1\rangle \end{aligned}$$

Therefore, $\phi = \frac{1}{4} = 0.01_2$

The measurement of q_1, q_0 yields the binary state $|01\rangle$, perfectly matching the binary fraction 0.01_2 .

Synthesis: The QPE & QFT cycle forms a complete encode-decode ecosystem

Encoding: Quantum Phase Estimation



Encoding: Quantum Phase Estimation

QPE leverages phase kickback to organically encode an unknown physical property (eigenvalues) into the continuous amplitudes of a quantum state.

Decoding: Inverse QFT



Decoding: Inverse QFT

The IQFT decodes this complex interference pattern, collapsing the phase data back into measurable, discrete binary decimals.

